Exercise Sheet 4

You may use the results of Lecture 4. These are the lifting theorem, the theorem on smooth consequences of covering groups, the proposition on discrete normal subgroups, Lie's integration theorem, the Descent Theorem and its corollary. Lie groups are as in the lecture. Exercises 4.1–4.6 are the core work. Exercise 4.7 is an optional extension.

Exercise 4.1. Discrete normal subgroups and connectedness. For $\theta \in \mathbb{R}$ put

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Every element of $\mathrm{O}(2)$ is R_θ or $R_\theta S$ for some θ. Let $N = \{R_0, R_{\pi/2}, R_\pi, R_{3\pi/2}\}$.

(a) Show that $SR_\theta S^{-1} = R_{-\theta}$. Deduce that N is a normal subgroup of $\mathrm{O}(2)$.

(b) Show that N is discrete but not central in $\mathrm{O}(2)$. Which hypothesis of the proposition on discrete normal subgroups fails? Prove that it fails.

(c) Let

$$D = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Show that $D, J \in \mathrm{SU}(2)$. Show that the only elements of $\mathrm{SU}(2)$ commuting with both D and J are $\pm I$. Deduce that every discrete normal subgroup of $\mathrm{SU}(2)$ lies in $\{\pm I\}$. You may use that $\mathrm{SU}(2)$ is connected.

Exercise 4.2. Scalar weights and periods. Fix $c > 0$ and let $p_c \colon \mathbb{R} \to \mathrm{U}(1)$, $p_c(t) = e^{2\pi i t/c}$. Use $\mathrm{Lie}\,\mathrm{U}(1) = i\mathbb{R}$, $\exp(i\theta) = e^{i\theta}$ and $\Lambda = \ker\exp = 2\pi i\mathbb{Z}$. For $\lambda \in \mathbb{C}$ let $\chi_\lambda(t) = e^{\lambda t}$.

(a) Show that p_c is a surjective smooth homomorphism with dp_c bijective. Compute $\ker p_c$.

(b) Let $\chi \colon \mathbb{R} \to \mathbb{C}^\times$ be a smooth homomorphism with $\chi'(0) = \lambda$. Show that $\chi = \chi_\lambda$ by solving an ODE.

(c) Find all λ for which some character χ of $\mathrm{U}(1)$ satisfies $\chi \circ p_c = \chi_\lambda$. Show that then $\chi(z) = z^n$ for some $n \in \mathbb{Z}$. Write $d\chi = i\ell$ and check that $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

(d) Take $c = 2\pi$. Decide the cases $\lambda = 2i$, $\lambda = i/3$ and $\lambda = 1 + i$. In each failed case display the contradiction.

Exercise 4.3. A finite quotient: $\mathrm{U}(2)$. Let $q \colon \mathrm{U}(1) \times \mathrm{SU}(2) \to \mathrm{U}(2)$, $q(z, g) = zg$. Let Π_m be the representation of $\mathrm{SU}(2)$ on V_m from Lecture 4, so that $\dim V_m = m + 1$ and $\Pi_m(-I) = (-1)^m I$. For $n \in \mathbb{Z}$ and $m \geq 0$ put $\Pi_{n,m}(z, g) = z^n \Pi_m(g)$.

(a) Show that q is a surjective smooth homomorphism with kernel $\{(1, I), (-1, -I)\}$ and bijective differential.

(b) Show that $\Pi_{n,m}$ is a representation. Say that $\Pi_{n,m}$ *descends through* q if some representation σ of $\mathrm{U}(2)$ satisfies $\sigma \circ q = \Pi_{n,m}$. Show that $\Pi_{n,m}$ descends through q if and only if $n \equiv m \pmod 2$.

(c) Identify the descended representations for $(n, m) = (1, 1)$ and $(2, 0)$. Show that $(1, 0)$ fails, with an explicit contradiction.

(d) The character $z \mapsto z$ of $\mathrm{U}(1)$ exists, and so does the trivial representation of $\mathrm{SU}(2)$. Explain why $(1, 0)$ still fails.

Exercise 4.4. **The covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ by hand.** Let $X_j = -\frac{i}{2}\sigma_j$ with the Pauli matrices of Lecture 4. They satisfy

$$\sigma_j^2 = I, \qquad \sigma_1\sigma_2 = i\sigma_3, \quad \sigma_2\sigma_3 = i\sigma_1, \quad \sigma_3\sigma_1 = i\sigma_2, \qquad \sigma_j\sigma_k = -\sigma_k\sigma_j \ (j \neq k).$$

Write $\mathrm{Ad}(g)$ as a matrix in the basis (X_1, X_2, X_3). Recall that $\mathrm{Ad}\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ is a covering map. Let

$$R_1(t) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos t & -\sin t \\ 0 & \sin t & \cos t \end{pmatrix}.$$

(a) Show that $\exp(tX_1) = \cos(t/2)\, I - i \sin(t/2)\, \sigma_1$, and display it as a matrix.

(b) Show that $\mathrm{Ad}(\exp tX_1) = R_1(t)$ by computing $u\sigma_2u^{-1}$ and $u\sigma_3u^{-1}$ for $u = \exp(tX_1)$.

(c) Let $\rho(t) = R_1(2\pi t)$ for $t \in [0,1]$. Find the lift of ρ starting at I. Show that it ends at $-I$, and conclude that ρ is not homotopic to a constant loop.

(d) Show that $\rho * \rho$ lifts to a loop, and that $\rho * \rho$ is homotopic to the constant loop.

Exercise 4.5. **Which spins descend.** Let V_m have basis $v_0, \dots, v_m$ with $\sigma_3 v_k = (m-2k)v_k$, and let Π_m be the representation of $\mathrm{SU}(2)$ integrating the restriction φ_m to $\mathfrak{su}(2)$. Let $X_3 = -\frac{i}{2}\sigma_3$. Recall that $\mathrm{Ad}(\exp \theta X_3) = R_3(\theta)$, the rotation by θ about the third axis. Put $\chi_m(\theta) = \operatorname{tr} \Pi_m(\exp \theta X_3)$.

(a) Show that

$$\chi_m(\theta) = \frac{\sin\big((m+1)\theta/2\big)}{\sin(\theta/2)} \qquad (\theta \notin 2\pi\mathbb{Z}).$$

(b) Show that $\chi_m(\theta + 2\pi) = (-1)^m \chi_m(\theta)$ and $\chi_m(2\pi) = (-1)^m(m+1)$.

(c) Let $m = 2j$, and let Π be the representation of $\mathrm{SO}(3)$ with $\Pi \circ \mathrm{Ad} = \Pi_m$. Show that $\operatorname{tr} \Pi(R_3(\theta)) = 1 + 2\sum_{r=1}^{j} \cos(r\theta)$. Compare the case $j = 1$ with $\operatorname{tr} R_3(\theta)$.

(d) Let $m = 3$. Show, using traces, that no representation Π of $\mathrm{SO}(3)$ has $d\Pi = \varphi_3 \circ (d\,\mathrm{Ad})^{-1}$.

Exercise 4.6. **What the differential does not see.** Use $\mathrm{O}(2)$ and S from Exercise 4.1, with Lie algebra $\mathfrak{so}(2)$ of real skew-symmetric 2×2 matrices. Use $\operatorname{Lie} \mathrm{U}(1) = i\mathbb{R}$ and the additive group $(\mathbb{R}, +)$, whose Lie algebra is $\mathbb{R}$ with $\exp = \mathrm{id}$. Let $p(t) = e^{it}$.

(a) Show that the trivial character and $\det$ of $\mathrm{O}(2)$ have the same differential. Explain why the uniqueness argument of the lecture does not apply.

(b) Let $\psi\colon i\mathbb{R} \to \mathbb{R}$, $\psi(i\theta) = \theta$. This is a Lie algebra homomorphism, since both algebras are abelian. Show in two ways that no smooth homomorphism $\Phi\colon \mathrm{U}(1) \to \mathbb{R}$ has $d\Phi = \psi$: first with the exponential at $\theta = 2\pi$, then with compact subgroups of $\mathbb{R}$.

(c) Integrate $\psi \circ dp$ on $\mathbb{R}$, and check that the descent criterion fails.

Exercise 4.7. **Optional extension: groups with Lie algebra $\mathfrak{su}(2)$.**

(a) Let G_1, G_2 be connected, simply connected Lie groups, and let $\alpha\colon \mathfrak{g}_1 \to \mathfrak{g}_2$ be a Lie algebra isomorphism. Show that there is a unique Lie group isomorphism $\Phi\colon G_1 \to G_2$ with $d\Phi = \alpha$.

(b) Let G be a connected Lie group whose Lie algebra is isomorphic to $\mathfrak{su}(2)$. Show that G is isomorphic to $\mathrm{SU}(2)$ or to $\mathrm{SO}(3)$.